

Distributed robust optimization under polyhedral uncertainty sets: a bilevel optimization approach^{*}

Chengrui Han^{*} Shu Liang^{**} Yue Zhao^{***} Yiheng Wei^{*,1}

^{*} School of Mathematics, Southeast University, Nanjing 211189, China
(e-mail: 220251992@seu.edu.cn, neudawei@seu.edu.cn)

^{**} School of Electronics and Information Engineering, Tongji University, Shanghai 201804, China (e-mail: sliang@tongji.edu.cn)

^{***} School of Mathematics and Statistics, Yunnan University, Kunming 650091, China (e-mail: zhaoyue@ynu.edu.cn)

Abstract: This paper investigates distributed robust consensus optimization for multi-agent systems under polyhedral uncertainty sets. Building on the theoretical connection between robust and bilevel optimization, we are the first to study this class of problems from a bilevel perspective and to establish a restatement of the original problem via strong duality in linear programming. We design a continuous-time distributed dynamical system based on the Karush–Kuhn–Tucker (KKT) optimality conditions, thereby providing a continuous-time counterpart to recent discrete-time algorithms. By constructing a Lyapunov function and invoking LaSalle’s invariance principle, we rigorously prove global asymptotic convergence. Numerical simulation verifies the effectiveness of the proposed approach.

Keywords: Distributed optimization, Robust optimization, Bilevel optimization, Polyhedral uncertainty sets

1. INTRODUCTION

Optimization under uncertainty is a classical topic in mathematical programming and industrial engineering. Its stochastic programming branch originated with Dantzig’s two-stage stochastic programming model (Dantzig, 1955), whereas robust optimization is rooted in Wald’s minimax criterion (Wald, 1945) and was systematically developed into an independent discipline by Ben-Tal and Nemirovski (1998; 1999; 2000). In recent years, with the proliferation of networked systems such as power grids and sensor networks (Yang et al., 2019; Boyd et al., 2011), distributed optimization has become increasingly important because of its scalability and robustness (Nedić and Ozdaglar, 2009; Nedić and Liu, 2018).

Distributed optimization can be broadly categorized into discrete-time and continuous-time approaches. Among discrete-time algorithms, Nedić et al.’s DGD algorithm converges slowly because it relies on diminishing step sizes (2009); subsequent fixed-step-size methods such as EXTRA (Shi et al., 2015) and DIGing (Nedić et al., 2017) improve performance through correction terms or gradient tracking. Continuous-time methods are typically designed using Lyapunov stability theory, including proportional-

integral dynamics by Wang and Elia (2010) and dynamics incorporating second-order Hessian information by Lu et al. (2011), the latter being better suited to practical dynamic scenarios such as robotics.

In distributed robust optimization, early studies focused on continuous-time approaches for general convex problems (Liu et al., 2017; Le et al., 2018) and on algorithms under polyhedral uncertainty sets (Zeng et al., 2018; Deng et al., 2023), while Yang et al. investigated equivalent transformations (2014). However, existing continuous-time approaches remain limited to specific problem structures. Motivated by recent discrete-time algorithms (Deng et al., 2023), this paper develops a continuous-time counterpart. Building on the theoretical connection between robust and bilevel optimization (Goerigk et al., 2025), we are the first to apply a bilevel reformulation to this class of problems and to establish rigorous convergence guarantees.

Contributions. This paper makes three main contributions: (1) We are the first to apply a bilevel optimization reformulation to distributed robust optimization under polyhedral uncertainty sets, a perspective absent from existing discrete-time approaches. Using strong duality in linear programming, we establish a restatement of the original problem. (2) We design a continuous-time distributed dynamical system based on the KKT optimality conditions. (3) We prove global asymptotic convergence using Lyapunov stability theory and LaSalle’s invariance principle. The remainder of the paper is organized as follows: Section 2 presents the preliminaries, Section 3 formulates and reconstructs the problem, Section 4 presents the

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¹ Corresponding author: Yiheng Wei (e-mail: neudawei@seu.edu.cn).

system design and convergence analysis, Section 5 provides numerical validation, and Section 6 concludes the paper.

2. PRELIMINARIES

Graph Theory. An undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has vertex set $\mathcal{V} = \{1, \dots, N\}$ and edge set $\mathcal{E}$. The Laplacian matrix is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D} = \text{diag}\{d_1, \dots, d_N\}$ and $\mathbf{A}$ is the adjacency matrix.

Lemma 1. (Laplacian kernel). (Gharesifard and Cortés, 2014) For a strongly connected graph with N vertices, $\text{Ker}(\mathbf{L}) = \text{span}\{\mathbf{1}_N\}$. Equivalently, $\mathbf{L}\mathbf{x} = \mathbf{0}$ if and only if $x_1 = \dots = x_N$.

Convex Optimization. A function $f : C \rightarrow \mathbb{R}$ on a convex set C is convex if $f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in C$ and $t \in [0, 1]$. The subdifferential is $\partial f(\mathbf{x}) = \{\boldsymbol{\xi} \mid f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \boldsymbol{\xi}, \mathbf{y} - \mathbf{x} \rangle, \forall \mathbf{y} \in C\}$, and the normal cone is $N_C(\mathbf{x}) = \{\mathbf{v} \mid \langle \mathbf{v}, \mathbf{y} - \mathbf{x} \rangle \leq 0, \forall \mathbf{y} \in C\}$.

Lemma 2. (KKT conditions). (Ruszczynski, 2006) Consider the convex optimization problem of the form:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, i = 1, \dots, \bar{m}, \\ & l_j(\mathbf{x}) = 0, j = 1, \dots, \bar{n}, \end{aligned}$$

where $\mathbf{x} \in C \subset \mathbb{R}^{\bar{d}}$, C is a closed convex set, f is convex, g_i is convex, and l_j is affine. If $\mathbf{x}^*$ is a local minimizer of the problem and the subdifferentials $\partial f(\mathbf{x}^*)$ and $\partial g_i(\mathbf{x}^*)$ exist, and Slater's condition holds, then there exist KKT multipliers λ_i, μ_j satisfying

$$\begin{cases} \mathbf{0} \in \partial f(\mathbf{x}^*) + \sum_{i=1}^{\bar{m}} \lambda_i \partial g_i(\mathbf{x}^*) + \sum_{j=1}^{\bar{n}} \mu_j \nabla l_j(\mathbf{x}^*) + N_C(\mathbf{x}^*), \\ g_i(\mathbf{x}^*) \leq 0, i = 1, \dots, \bar{m}, \\ l_j(\mathbf{x}^*) = 0, j = 1, \dots, \bar{n}, \\ \lambda_i \geq 0, i = 1, \dots, \bar{m}, \\ \lambda_i g_i(\mathbf{x}^*) = 0, i = 1, \dots, \bar{m}, \end{cases}$$

where ∂ denotes the subdifferential, ∇ denotes the gradient, and $N_C(\mathbf{x}^*)$ is the normal cone.

Projection Operators. For a closed convex set C , the projection operator is defined as:

$$P_C(\mathbf{x}) = \arg \min_{\mathbf{y} \in C} \|\mathbf{x} - \mathbf{y}\|_2.$$

Lemma 3. (Projection properties). (Deng et al., 2020) For a closed convex set C : (1) $\langle \mathbf{x} - P_C(\mathbf{x}), P_C(\mathbf{x}) - \mathbf{y} \rangle \geq 0$ for all $\mathbf{y} \in C$; (2) $\|P_C(\mathbf{x}) - P_C(\mathbf{y})\|_2^2 \leq \langle P_C(\mathbf{x}) - P_C(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$; (3) $\mathbf{x} = P_C(\mathbf{z})$ if and only if $\mathbf{z} - \mathbf{x} \in N_C(\mathbf{x})$.

Stability Theory. For a differential inclusion $\dot{\mathbf{x}}(t) \in F(\mathbf{x}(t))$, the following principle provides convergence guarantees.

Lemma 4. (LaSalle's principle). (Aubin and Cellina, 1984) Let the mapping $F : \mathbb{R}^{\bar{d}} \rightarrow \mathcal{B}(\mathbb{R}^{\bar{d}})$ be locally bounded, upper semicontinuous, and take nonempty compact convex values, and $V : \mathbb{R}^{\bar{d}} \rightarrow \mathbb{R}$ be a regular, locally Lipschitz function. Let K be a compact positively invariant set, $S = \{\mathbf{x} \in \mathbb{R}^{\bar{d}} \mid 0 \in L_F V(\mathbf{x})\}$, and $\bar{S}$ be the closure of S . If for all $\mathbf{x} \in K$, $\max L_F V(\mathbf{x}) \leq 0$, then any solution starting from K converges to the largest weakly invariant set $M \subseteq K \cap \bar{S}$.

3. PROBLEM FORMULATION AND RECONSTRUCTION

Consider a multi-agent system with N agents. Each agent i has local objective $f_i(\mathbf{x}_i)$ and optimization variable $\mathbf{x}_i \in \Omega_i \subset \mathbb{R}^n$. We study the following distributed robust consensus optimization problem:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \sum_{i=1}^N f_i(\mathbf{x}_i) \\ \text{s.t.} \quad & \mathbf{x}_i = \mathbf{x}_j \quad (\forall i, j \in \{1, \dots, N\}), \\ & \mathbf{g}_i(\mathbf{x}_i) \leq \mathbf{0}, \\ & \mathbf{A}_i \mathbf{x}_i - \mathbf{b}_i = \mathbf{0}, \\ & \mathbf{r}_i^\top \mathbf{x}_i \leq c_i \quad (\forall \mathbf{r}_i \in \mathcal{U}_i, 1 \leq i \leq N), \\ & \mathcal{U}_i := \{\mathbf{r}_i \mid \mathbf{D}_i \mathbf{r}_i \leq \mathbf{e}_i\}, \\ & \mathbf{x} \in \Omega, \end{aligned} \tag{1}$$

where $\mathbf{x}_i \in \Omega_i \subset \mathbb{R}^n$, Ω_i is a closed convex set, $\mathbf{x} = \text{col}\{\mathbf{x}_i\}_{i=1}^N$, $\Omega = \Omega_1 \times \dots \times \Omega_N$, $\mathbf{L}_n = \mathbf{L} \otimes \mathbf{I}_n$, $\mathbf{I}_n$ is the $n \times n$ identity matrix, $f_i : \Omega_i \rightarrow \mathbb{R}$, $\mathbf{g}_i : \Omega_i \rightarrow \mathbb{R}^{p_i}$, $\mathbf{A}_i \in \mathbb{R}^{m_i \times n}$, $\mathbf{b}_i \in \mathbb{R}^{m_i}$, $\mathbf{r}_i \in \mathbb{R}^n$, $c_i \in \mathbb{R}$, $\mathbf{D}_i \in \mathbb{R}^{q_i \times n}$, $\mathbf{e}_i \in \mathbb{R}^{q_i}$. Let $p = \sum_{i=1}^N p_i$, $m = \sum_{i=1}^N m_i$, $q = \sum_{i=1}^N q_i$.

Assumption 1: The network topology is an undirected connected graph.

Assumption 2: The objective function f_i and inequality constraint $\mathbf{g}_i$ of each agent are convex on an open set containing the closed convex set Ω_i .

Assumption 3: Slater's condition holds for problem (1).

Assumption 4: Problem (1) has an optimal solution.

Assumption 5: Each uncertainty set $\mathcal{U}_i$ is nonempty and compact.

3.1 Robust Optimization

A robust optimization problem is generally formulated as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & H(\mathbf{x}) \\ \text{s.t.} \quad & h_i(\mathbf{x}, \mathbf{u}_i) \leq \mathbf{0} \quad \forall i \in I, \forall \mathbf{u}_i \in \mathbf{U}_i, \\ & h_j(\mathbf{x}) \leq \mathbf{0} \quad \forall j \in J, \end{aligned} \tag{2}$$

where $I, J \subset \mathbb{N}$, $I \cap J = \emptyset$, $|I| < +\infty$, $|J| < +\infty$, $H : \mathbb{R}^{d_x} \rightarrow \mathbb{R}$, $h_i : \mathbb{R}^{d_x} \times \mathbb{R}^{d_{u_i}} \rightarrow \mathbb{R}^{\hat{m}_i}$, $h_j : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{\hat{n}_j}$, $\mathbf{U}_i$ is a bounded set defined by finitely many constraints, called the uncertainty set.

Let $\mathbf{x} \in \mathbb{R}^{d_x}$ be a feasible point for problem (2), meaning that $\mathbf{x}$ satisfies the constraints of problem (2). Let $\mathbf{x}^*$ be an optimal solution to problem (2), meaning that $\mathbf{x}^*$ is a feasible point that minimizes H .

If the uncertainty set $\mathbf{U}_i$ depends on the optimization variable $\mathbf{x}$, then problem (2) is a decision-dependent robust optimization problem.

3.2 Bilevel Optimization

A bilevel optimization problem is generally formulated as follows:

$$\begin{aligned}
& \min_{\mathbf{x}} F(\mathbf{x}, \mathbf{y}) \\
& \text{s.t. } G(\mathbf{x}, \mathbf{y}) \leq \mathbf{0}, \\
& \quad \mathbf{y} \in S(\mathbf{x}),
\end{aligned} \tag{3}$$

where $S(\mathbf{x})$ is the optimal solution set of the lower-level optimization problem parameterized by $\mathbf{x}$:

$$\begin{aligned}
& \min_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}) \\
& \text{s.t. } g(\mathbf{x}, \mathbf{y}) \leq \mathbf{0}.
\end{aligned} \tag{4}$$

Problem (3) is the upper-level optimization problem, and problem (4) is the lower-level optimization problem, where $F : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$, $G : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}^{\hat{m}}$, $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$, $g : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}^{\hat{n}}$.

If the lower-level problem does not have a unique optimal solution, then problem (3) is ill-defined. To address this ambiguity, we introduce the optimistic and pessimistic variants of bilevel optimization.

The optimistic bilevel optimization problem takes the form:

$$\begin{aligned}
& \min_{\mathbf{x}, \mathbf{y}} F(\mathbf{x}, \mathbf{y}) \\
& \text{s.t. } G(\mathbf{x}, \mathbf{y}) \leq \mathbf{0}, \\
& \quad \mathbf{y} \in S(\mathbf{x}).
\end{aligned} \tag{5}$$

The pessimistic bilevel optimization problem takes the form:

$$\begin{aligned}
& \min_{\mathbf{x}} \max_{\mathbf{y} \in S(\mathbf{x})} F(\mathbf{x}, \mathbf{y}) \\
& \text{s.t. } G(\mathbf{x}, \mathbf{y}') \leq \mathbf{0} \quad \forall \mathbf{y}' \in S(\mathbf{x}).
\end{aligned} \tag{6}$$

Let $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$ be a feasible point for problem (3), meaning that $(\mathbf{x}, \mathbf{y})$ satisfies the constraints of the upper-level problem and $\mathbf{y}$ is an optimal solution to the lower-level problem for the given $\mathbf{x}$. Let $(\mathbf{x}^*, \mathbf{y}^*)$ be a bilevel optimal solution to problem (3), meaning that $(\mathbf{x}^*, \mathbf{y}^*)$ is a feasible point that minimizes F .

Remark 5. Bilevel optimization provides a more general framework than robust optimization (Goerigk et al., 2025). When the uncertainty set is polyhedral and strong duality holds, the bilevel reformulation reduces to the standard robust counterpart (Zeng et al., 2018); this equivalence may fail for non-convex or decision-dependent uncertainty. Moreover, the bilevel formulation admits KKT-based reformulations that facilitate distributed algorithm design.

3.3 Problem Reconstruction

Under Assumption 1 and Lemma 1, problem (1) can be rewritten as the following bilevel optimization problem:

$$\begin{aligned}
& \min_{\mathbf{x}} f(\mathbf{x}) \\
& \text{s.t. } \mathbf{L}_n \mathbf{x} = \mathbf{0}, \\
& \quad \mathbf{g}_i(\mathbf{x}_i) \leq \mathbf{0}, \\
& \quad \mathbf{A}_i \mathbf{x}_i = \mathbf{b}_i, \\
& \quad \mathbf{x} \in \Omega, \\
& \quad \mathbf{r}_i^\top \mathbf{x}_i \leq c_i, \\
& \quad \mathbf{r}_i \in S_i(\mathbf{x}_i),
\end{aligned} \tag{7}$$

where $S_i(\mathbf{x}_i)$ is the optimal solution set of the lower-level optimization problem parameterized by $\mathbf{x}_i$:

$$\begin{aligned}
& \sup_{\mathbf{r}_i} \mathbf{r}_i^\top \mathbf{x}_i \\
& \text{s.t. } \mathbf{D}_i \mathbf{r}_i \leq \mathbf{e}_i.
\end{aligned} \tag{8}$$

Remark 6. The upper-level constraint depends only on the optimal value $v_i(\mathbf{x}_i) := \sup_{\mathbf{r}_i \in \mathcal{U}_i} \mathbf{r}_i^\top \mathbf{x}_i$, rather than on a particular optimal solution $\mathbf{r}_i \in S_i(\mathbf{x}_i)$. Since all optimal solutions yield the same value, the condition $\mathbf{r}_i^\top \mathbf{x}_i \leq c_i$ is equivalent to $v_i(\mathbf{x}_i) \leq c_i$. Therefore, the optimistic and pessimistic bilevel formulations of (7) coincide.

For any given $\mathbf{x}$, problem (8) is a linear program. Under Assumption 5, $\mathcal{U}_i$ is nonempty and compact, so the lower-level LP (8) is feasible and bounded; hence strong duality holds. Consequently, the constraint can be equivalently reformulated as:

$$\begin{aligned}
& \inf_{\mathbf{z}_i} \mathbf{z}_i^\top \mathbf{e}_i \\
& \text{s.t. } \mathbf{D}_i^\top \mathbf{z}_i = \mathbf{x}_i, \\
& \quad \mathbf{z}_i \geq \mathbf{0}.
\end{aligned} \tag{9}$$

Since the upper-level constraint depends only on the optimal value of the lower-level problem, the bilevel formulation can be expressed through the value function of the lower-level problem. By invoking strong duality for the linear lower-level problem (8), the value function can be represented by the dual problem (9). Consequently, problem (7) can be reconstructed as the following single-level optimization problem:

$$\begin{aligned}
& \min_{\mathbf{x}, \mathbf{z}} f(\mathbf{x}) \\
& \text{s.t. } \mathbf{L}_n \mathbf{x} = \mathbf{0}, \\
& \quad \mathbf{g}_i(\mathbf{x}_i) \leq \mathbf{0}, \\
& \quad \mathbf{A}_i \mathbf{x}_i = \mathbf{b}_i, \\
& \quad \mathbf{z}_i^\top \mathbf{e}_i \leq c_i, \\
& \quad \mathbf{D}_i^\top \mathbf{z}_i = \mathbf{x}_i, \\
& \quad \mathbf{x} \in \Omega,
\end{aligned} \tag{10}$$

where $\mathbf{z}_i \in \mathbb{R}_+^{q_i}$, $\mathbf{z} = \text{col}\{\mathbf{z}_i\}_{i=1}^N$.

The following lemma guarantees that any optimal solution to problem (10) is also an optimal solution to problem (1), justifying the study of problem (10) as a valid reconstruction of the original problem.

Lemma 7. ((Goerigk et al., 2025)). For each feasible point $\mathbf{x}$ of problem (2), if for each $i \in I$, the objective function $h_i(\mathbf{x}_i, \cdot)$ is continuous and the constraint set $\mathcal{U}_i$ is nonempty and compact, then problem (5) is a restatement of problem (2).

Remark 8. A restatement from problem A to problem B means that there exists a reconstruction independent of the objective function such that solving problem B yields a solution to problem A. This general theoretical result (Goerigk et al., 2025) provides the foundation for our specific application to distributed robust optimization under polyhedral uncertainty sets.

Proposition 9. Problem (7) is a restatement of problem (1).

Proof. Problem (1) is a static robust optimization problem, and problem (7) is the corresponding optimistic bilevel optimization problem. Since $\mathbf{r}_i^\top \mathbf{x}_i - c_i$ is continuous in $\mathbf{r}_i$ for any fixed $\mathbf{x}_i$, and $\mathcal{U}_i$ is nonempty and compact by

Assumption 5, by Lemma 7, problem (7) is a restatement of problem (1).

Remark 10. By the strong duality reduction above, problem (10) provides a single-level reconstruction of problem (7); hence any optimal solution to problem (10) is also an optimal solution to problem (1).

4. DISTRIBUTED DYNAMICAL SYSTEM DESIGN

We introduce the following notation for the system design. For each agent i , define

$$\varphi(\mathbf{x}_i) = \sum_{j \in \mathcal{N}_i} a_{ij}(\mathbf{x}_i - \mathbf{x}_j),$$

where $\mathcal{N}_i$ denotes the neighborhood of agent i , and a_{ij} is the element in the i -th row and j -th column of the adjacency matrix.

Under Assumption 2, problem (10) is a convex optimization problem. Under Assumption 3, Slater's condition holds, so by Lemma 2, the KKT conditions are necessary and sufficient for the optimality of problem (10). Based on these conditions, we propose the following continuous-time distributed dynamical system:

$$\begin{cases} \dot{\mathbf{y}}_i = \mathbf{x}_i - u_i(\mathbf{x}_i) - \varphi(\boldsymbol{\alpha}_i) - v_i(\mathbf{x}_i)^\top \tilde{\boldsymbol{\beta}}_i \\ \quad - \mathbf{A}_i^\top \boldsymbol{\gamma}_i + \boldsymbol{\lambda}_i - \mathbf{y}_i, \\ \dot{\mathbf{z}}_i = -\mathbf{z}_i + \tilde{\mathbf{z}}_i - \mathbf{e}_i \tilde{\boldsymbol{\eta}}_i - \mathbf{D}_i \boldsymbol{\lambda}_i, \\ \dot{\boldsymbol{\alpha}}_i = \varphi(\mathbf{x}_i), \\ \dot{\boldsymbol{\beta}}_i = -\boldsymbol{\beta}_i + \tilde{\boldsymbol{\beta}}_i + \mathbf{g}_i(\mathbf{x}_i), \\ \dot{\boldsymbol{\gamma}}_i = \mathbf{A}_i \mathbf{x}_i - \mathbf{b}_i, \\ \dot{\boldsymbol{\eta}}_i = -\boldsymbol{\eta}_i + \tilde{\boldsymbol{\eta}}_i + \mathbf{e}_i^\top \tilde{\mathbf{z}}_i - \mathbf{c}_i, \\ \dot{\boldsymbol{\lambda}}_i = \mathbf{D}_i^\top \tilde{\mathbf{z}}_i - \mathbf{x}_i, \\ \mathbf{x}_i = P_{\Omega_i}(\mathbf{y}_i), \end{cases} \quad (11)$$

where $u_i(\mathbf{x}_i) \in \partial f_i(\mathbf{x}_i)$, $v_i(\mathbf{x}_i) \in \partial \mathbf{g}_i(\mathbf{x}_i)$, $\tilde{\mathbf{z}}_i = P_{\mathbb{R}_+^{q_i}}(\mathbf{z}_i)$, $\tilde{\boldsymbol{\beta}}_i = P_{\mathbb{R}_+^{p_i}}(\boldsymbol{\beta}_i)$, $\tilde{\boldsymbol{\eta}}_i = P_{\mathbb{R}_+^n}(\boldsymbol{\eta}_i)$.

For subsequent analysis, we rewrite system (11) in compact form:

$$\begin{cases} \dot{\mathbf{y}} = \mathbf{x} - \mathbf{u}(\mathbf{x}) - \mathbf{L}_n \boldsymbol{\alpha} - \mathbf{v}(\mathbf{x})^\top \tilde{\boldsymbol{\beta}} \\ \quad - \mathbf{A}^\top \boldsymbol{\gamma} + \boldsymbol{\lambda} - \mathbf{y}, \\ \dot{\mathbf{z}} = -\mathbf{z} + \tilde{\mathbf{z}} - \mathbf{e} \tilde{\boldsymbol{\eta}} - \mathbf{D} \boldsymbol{\lambda}, \\ \dot{\boldsymbol{\alpha}} = \mathbf{L}_n \mathbf{x}, \\ \dot{\boldsymbol{\beta}} = -\boldsymbol{\beta} + \tilde{\boldsymbol{\beta}} + \mathbf{g}(\mathbf{x}), \\ \dot{\boldsymbol{\gamma}} = \mathbf{A} \mathbf{x} - \mathbf{b}, \\ \dot{\boldsymbol{\eta}} = -\boldsymbol{\eta} + \tilde{\boldsymbol{\eta}} + \mathbf{e}^\top \tilde{\mathbf{z}} - \mathbf{c}, \\ \dot{\boldsymbol{\lambda}} = \mathbf{D}^\top \tilde{\mathbf{z}} - \mathbf{x}, \\ \mathbf{x} = P_\Omega(\mathbf{y}), \end{cases} \quad (12)$$

where $\mathbf{x} = \text{col}\{\mathbf{x}_i\}_{i=1}^N \in \mathbb{R}^{Nn}$, $\mathbf{y} = \text{col}\{\mathbf{y}_i\}_{i=1}^N \in \mathbb{R}^{Nn}$, $\mathbf{z} = \text{col}\{\mathbf{z}_i\}_{i=1}^N \in \mathbb{R}^q$, $\tilde{\mathbf{z}} = \text{col}\{\tilde{\mathbf{z}}_i\}_{i=1}^N \in \mathbb{R}^q$, $\mathbf{u}(\mathbf{x}) = \text{col}\{u_i(\mathbf{x}_i)\}_{i=1}^N$, $\mathbf{v}(\mathbf{x}) = \text{col}\{v_i(\mathbf{x}_i)\}_{i=1}^N$, $\mathbf{g}(\mathbf{x}) = \text{col}\{\mathbf{g}_i(\mathbf{x}_i)\}_{i=1}^N$, $\boldsymbol{\alpha} = \text{col}\{\boldsymbol{\alpha}_i\}_{i=1}^N \in \mathbb{R}^{Nn}$, $\boldsymbol{\beta} = \text{col}\{\boldsymbol{\beta}_i\}_{i=1}^N \in \mathbb{R}^p$, $\tilde{\boldsymbol{\beta}} = \text{col}\{\tilde{\boldsymbol{\beta}}_i\}_{i=1}^N$, $\boldsymbol{\gamma} = \text{col}\{\boldsymbol{\gamma}_i\}_{i=1}^N \in \mathbb{R}^m$, $\boldsymbol{\eta} = \text{col}\{\boldsymbol{\eta}_i\}_{i=1}^N \in \mathbb{R}^n$, $\tilde{\boldsymbol{\eta}} = \text{col}\{\tilde{\boldsymbol{\eta}}_i\}_{i=1}^N$, $\boldsymbol{\lambda} = \text{col}\{\boldsymbol{\lambda}_i\}_{i=1}^N \in \mathbb{R}^{Nn}$, $\mathbf{A} = \text{diag}\{\mathbf{A}_i\}_{i=1}^N$, $\mathbf{b} = \text{col}\{\mathbf{b}_i\}_{i=1}^N$, $\mathbf{c} = \text{col}\{\mathbf{c}_i\}_{i=1}^N$, $\mathbf{D} = \text{diag}\{\mathbf{D}_i\}_{i=1}^N$, $\mathbf{e} = \text{diag}\{\mathbf{e}_i\}_{i=1}^N$.

4.1 Optimality and Equilibrium Equivalence

Proposition 11. If $(\mathbf{y}^*, \mathbf{z}^*, \boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*, \boldsymbol{\eta}^*, \boldsymbol{\lambda}^*) \in \mathbb{R}^{Nn} \times \mathbb{R}^q \times \mathbb{R}^{Nn} \times \mathbb{R}^p \times \mathbb{R}^m \times \mathbb{R}^N \times \mathbb{R}^{Nn}$ is an equilibrium point of dynamical system (12), then $(\mathbf{x}^*, \tilde{\mathbf{z}}^*)$ is an optimal solution to problem (10). Conversely, if $(\mathbf{x}^*, \tilde{\mathbf{z}}^*) \in \Omega \times \mathbb{R}_+^q$ is an optimal solution to problem (10), then there exist $(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*, \boldsymbol{\eta}^*, \boldsymbol{\lambda}^*)$ such that $(\mathbf{y}^*, \mathbf{z}^*, \boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*, \boldsymbol{\eta}^*, \boldsymbol{\lambda}^*)$ is an equilibrium point of dynamical system (12).

Remark 12. The proof follows directly from the KKT optimality conditions (Lemma 2) and is omitted due to space constraints.

4.2 Convergence Analysis

Theorem 13. (Global asymptotic convergence). Under Assumptions 1–5, any trajectory of system (12) starting from an arbitrary initial state converges to an equilibrium point that corresponds to an optimal solution to problem (10).

Proof. By Assumption 4 and Proposition 11, let $(\mathbf{x}^*, \tilde{\mathbf{z}}^*)$ be an optimal solution to problem (10). Then there exist $(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*, \boldsymbol{\eta}^*, \boldsymbol{\lambda}^*)$ such that

$$(\mathbf{y}^*, \mathbf{z}^*, \boldsymbol{\alpha}^*, \boldsymbol{\beta}^*, \boldsymbol{\gamma}^*, \boldsymbol{\eta}^*, \boldsymbol{\lambda}^*)$$

is an equilibrium point of dynamical system (12), and satisfies

$$\begin{aligned} \mathbf{x}^* &= P_\Omega(\mathbf{y}^*), \quad \tilde{\mathbf{z}}^* = P_{\mathbb{R}_+^q}(\mathbf{z}^*), \\ \tilde{\boldsymbol{\beta}}^* &= P_{\mathbb{R}_+^p}(\boldsymbol{\beta}^*), \quad \tilde{\boldsymbol{\eta}}^* = P_{\mathbb{R}_+^n}(\boldsymbol{\eta}^*). \end{aligned}$$

Consider the following Lyapunov function:

$$\begin{aligned} V(\mathbf{y}) &= \frac{1}{2} (\|\mathbf{y} - P_\Omega(\mathbf{y}^*)\|^2 - \|\mathbf{y} - P_\Omega(\mathbf{y})\|^2), \\ V(\mathbf{z}) &= \frac{1}{2} (\|\mathbf{z} - \tilde{\mathbf{z}}^*\|^2 - \|\mathbf{z} - \tilde{\mathbf{z}}\|^2), \\ V(\boldsymbol{\alpha}) &= \frac{1}{2} \|\boldsymbol{\alpha} - \boldsymbol{\alpha}^*\|^2, \\ V(\boldsymbol{\beta}) &= \frac{1}{2} (\|\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}^*\|^2 - \|\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}\|^2), \\ V(\boldsymbol{\gamma}) &= \frac{1}{2} \|\boldsymbol{\gamma} - \boldsymbol{\gamma}^*\|^2, \\ V(\boldsymbol{\eta}) &= \frac{1}{2} (\|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}^*\|^2 - \|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}\|^2), \\ V(\boldsymbol{\lambda}) &= \frac{1}{2} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|^2. \end{aligned}$$

Then the Lyapunov function is

$$V(\mathbf{y}, \mathbf{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \boldsymbol{\lambda}) = V(\mathbf{y}) + V(\mathbf{z}) + V(\boldsymbol{\alpha}) + V(\boldsymbol{\beta}) + V(\boldsymbol{\gamma}) + V(\boldsymbol{\eta}) + V(\boldsymbol{\lambda}),$$

denoted by $V(t)$.

By Lemma 3,

$$\begin{aligned} V(t) &\geq \frac{1}{2} (\|\mathbf{x} - \mathbf{x}^*\|^2 + \|\tilde{\mathbf{z}} - \tilde{\mathbf{z}}^*\|^2 + \|\boldsymbol{\alpha} - \boldsymbol{\alpha}^*\|^2 \\ &\quad + \|\tilde{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}^*\|^2 + \|\boldsymbol{\gamma} - \boldsymbol{\gamma}^*\|^2 \\ &\quad + \|\tilde{\boldsymbol{\eta}} - \tilde{\boldsymbol{\eta}}^*\|^2 + \|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|^2) \geq 0, \end{aligned}$$

hence $V(t)$ is positive definite and radially unbounded.

For the differential inclusion system (12), let $L_F V$ denote the set-valued Lie derivative of V along F . For any $l \in L_F V$, there exist $u(\mathbf{x}) \in \partial f(\mathbf{x})$ and $v(\mathbf{x}) \in \partial \mathbf{g}(\mathbf{x})$ such that $l = \dot{V}_y + \dot{V}_z + \dot{V}_\alpha + \dot{V}_\beta + \dot{V}_\gamma + \dot{V}_\eta + \dot{V}_\lambda$. By the projection

properties in Lemma 3 and the convexity of the objective functions in Assumption 2, it follows that $l \leq 0$, and hence $\max L_F V \leq 0$.

Since $\max L_F V \leq 0$, $V(t)$ is non-increasing along any trajectory of system (12), so the trajectory remains in the level set:

$$K = \{(\mathbf{y}, \mathbf{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \boldsymbol{\lambda}) \mid V(t) \leq V(0)\}$$

for all $t \geq 0$, i.e., K is positively invariant. Since $V(t)$ is positive definite and radially unbounded, K is also compact. Let

$$S = \{(\mathbf{y}, \mathbf{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \boldsymbol{\lambda}) \mid 0 \in L_F V\},$$

and let M be the largest weakly invariant set in $K \cap \bar{S}$, which is nonempty. By Lemma 4, $(\mathbf{y}, \mathbf{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \boldsymbol{\lambda})$ converges to M . Therefore,

$$\lim_{t \rightarrow +\infty} (\mathbf{x}(t), \tilde{\mathbf{z}}(t)) = (\mathbf{x}^*, \tilde{\mathbf{z}}^*).$$

Remark 14. By combining Proposition 9 and Theorem 13, we establish that the distributed system (11) globally asymptotically converges to an equilibrium point corresponding to an optimal solution to the original problem (1). This result is achieved through the bilevel optimization reformulation, which we apply for the first time to distributed robust optimization under polyhedral uncertainty sets.

5. NUMERICAL SIMULATION

To validate the effectiveness of the proposed distributed dynamical system (11), we present a numerical simulation implemented in Python using a Runge-Kutta integrator. Consider a four-agent system with objectives f_i , $i = 1, \dots, 4$, as specified in Table 1. The network topology is shown in Fig. 1.

Table 1. Objective functions for agents.

Agent	Objective Function
f_1	$3x_{11} - 2x_{12}$
f_2	$x_{21}^2 + x_{22}^2$
f_3	$ x_{31} + x_{32} $
f_4	$\max(0, x_{41}) + \max(0, x_{42})$

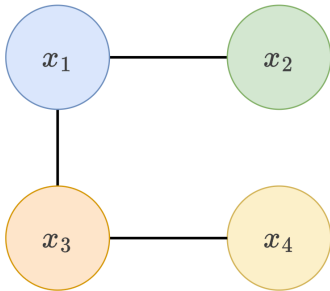


Fig. 1. Multi-agent network topology.

The Laplacian matrix $\mathbf{L}$ has the form:

$$\mathbf{L} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}.$$

Each agent's decision variable satisfies $\mathbf{x}_i \in \mathbb{R}^2$, with local constraint set $\Omega_i = \{\mathbf{x}_i \in \mathbb{R}^2 \mid \|\mathbf{x}_i\|_2 \leq 5\}$. The

uncertainty matrices are $\mathbf{D}_i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ for $1 \leq i \leq 4$, and c_i and $\mathbf{e}_i$ are given in Table 2.

Table 2. Uncertainty set parameters.

	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$
c_i	10	12	12	16
$\mathbf{e}_i^\top$	[1 1]	[2 1]	[1 2]	[2 2]

It is readily verified that the above optimization problem satisfies Assumptions 1–5 in Section 3, and the theoretical optimal solution is $\mathbf{x}_i^* = [0 \ 0]^\top$ for all $i = 1, \dots, 4$.

Figures 2 and 3 illustrate the convergence performance. Both the consensus error $\|\mathbf{x} - \mathbf{x}^*\|_2$ and the objective-value error $|f(\mathbf{x}) - f(\mathbf{x}^*)|$ converge to zero asymptotically, validating the effectiveness of the proposed continuous-time distributed approach.

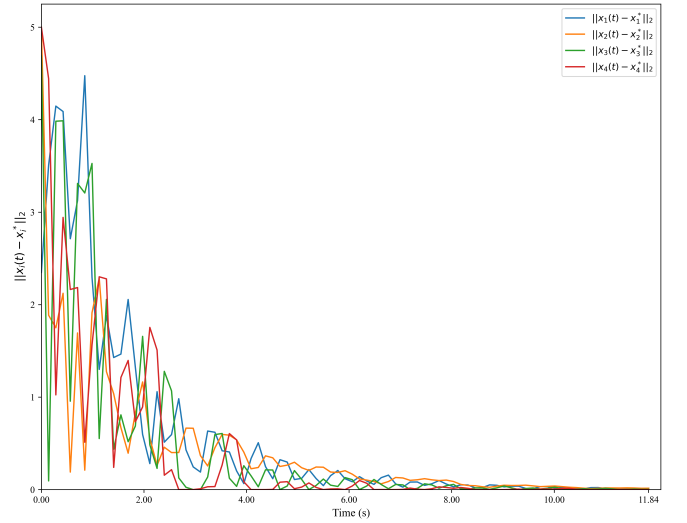


Fig. 2. Convergence of the consensus error $\|\mathbf{x} - \mathbf{x}^*\|_2$.

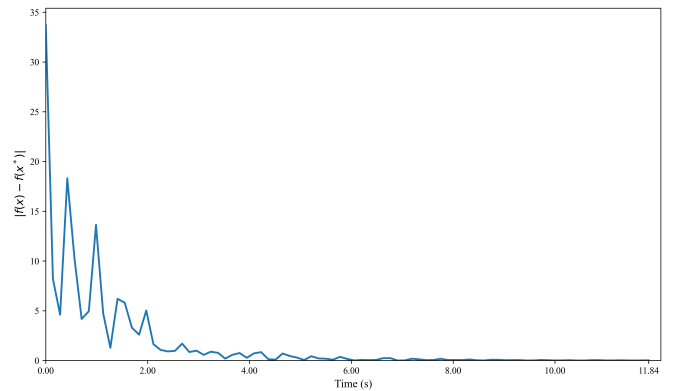


Fig. 3. Convergence of the objective-value error $|f(\mathbf{x}) - f(\mathbf{x}^*)|$.

6. CONCLUSION

This paper addresses distributed robust optimization for multi-agent systems under polyhedral uncertainty sets. Building on the theoretical connection between robust

and bilevel optimization, we employ a bilevel reformulation and strong duality in linear programming to establish a restatement of the original problem. We design a continuous-time distributed dynamical system based on the KKT optimality conditions and rigorously prove global asymptotic convergence using Lyapunov stability theory and LaSalle's invariance principle. Numerical simulation demonstrates the effectiveness of the proposed approach. Future research directions include extensions to time-varying network topologies, communication delays, and non-convex objective functions.

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